

Software Engineering Project

Jan 2026 Term - Team 51

Milestone - 1



Submitted by:

Kaushik Gaur (23f1000687)

Shriyanshi Parashari (22f3001012)

Aadil Iqbal (23f1001130)

Nachiket Gore (23f1000542)

IDENTIFICATION OF USERS

Primary User

1. Restaurant Manager (Café Owner / Operations Manager)

The main decision-maker and controller of daily operations. Responsible for:

- Managing menu items
- Defining ingredient requirements
- Monitoring inventory
- Approving customer orders
- Controlling pricing and profit margins
- Overseeing workflow between customer and kitchen

This role directly interacts with most core functionalities of the system.

Secondary Users

2. Customers (Dine-in Users)

Customers interact with the system by:

- Scanning QR codes at tables
- Viewing digital menus
- Placing orders
- Requesting custom dishes
- Checking dish availability

They expect fast, accurate service and transparency in ordering.

3. Chef / Kitchen Staff

Responsible for:

- Viewing approved orders
- Preparing dishes
- Marking orders as completed
- Following structured preparation instructions

They rely on clarity, order sequencing, and accurate information.

Tertiary User

4. Waiter (Service Staff)

The waiter does not directly interact with the software interface but is operationally affected by it.

The waiter is responsible for:

- Delivering prepared dishes to the correct tables
- Receiving notification when an order is ready
- Coordinating between kitchen and customers during service

Since the waiter does not operate the system directly but depends on accurate order status and table tagging generated by the system, this role is classified as a tertiary user.

2. User Research – Pain Points Identified

Based on interviews and observation studies conducted with restaurant staff and customers, the following recurring problems were identified:

Manager Pain Points

- Difficulty tracking real-time inventory during busy hours
 - Selling dishes that later turn out to be unavailable
 - Manual coordination between service and kitchen
 - Lack of structured workflow for order approval
 - No systematic pricing logic for custom requests
-

Customer Pain Points

- Receiving wrong-table deliveries
- Confusion during peak hours
- Not knowing whether dishes are actually available
- Limited structured customization options
- Delays caused by manual order taking

Chef Pain Points

- Receiving incomplete or unclear order instructions
 - Confusion during rush hours
 - Lack of order prioritization visibility
 - Verbal miscommunication with service staff
-

Waiter Pain Points

- Delivering food to wrong tables due to unclear communication
- Not knowing when an order is fully ready
- Miscommunication between kitchen and service staff
- Handling customer dissatisfaction caused by order errors

These insights guided the formulation of structured user stories below.

3. User Stories

Manager Stories

Story 1

As a Restaurant Manager,

I want the system to automatically deduct ingredient quantities immediately after an order is approved,

So that inventory records remain accurate at all times during operational hours.

Story 2

As a Restaurant Manager,

I want dishes to automatically become unavailable on the menu when required ingredients reach zero stock,

So that customers cannot order items that cannot be prepared.

Story 3

As a Restaurant Manager,

I want to define ingredient cost and profit margin for AI-generated custom dishes,

So that final pricing is calculated accurately before customer confirmation.

Story 4

As a Restaurant Manager,

I want to review and approve customer orders digitally before they reach the kitchen,
So that operational control and accountability are maintained.

Customer Stories

Story 5

As a Customer,

I want to scan a unique QR code at my table and place orders digitally,
So that my order is correctly tagged to my table and delivered accurately.

Story 6

As a Customer,

I want to request dish suggestions based on dietary preferences (e.g., high-protein or low-calorie),
So that I can choose meals aligned with my requirements.

Story 7

As a Customer,

I want to see the final price of a custom-generated dish before confirming the order,
So that I can make an informed purchasing decision.

Chef Stories

Story 8

As a Chef,

I want to view approved orders in a structured queue format,
So that I can prioritize and manage preparation efficiently during peak hours.

Story 9

As a Chef,

I want to mark orders as completed once preparation is finished,
So that the system triggers the next operational step without confusion.

Story 10

As a Chef,

I want clear dish details and quantities displayed for every approved order,
So that preparation errors are minimized.

Waiter Story (Tertiary User)

Story 11

As a Waiter,

I want to receive clear notification when an order is marked as ready,

So that I can deliver the correct dish to the correct table without confusion.

Interview with the owner - [Video](#)
